

MDSim: Neural Multi-Layer Draped Garment Simulation with Differentiable Friction

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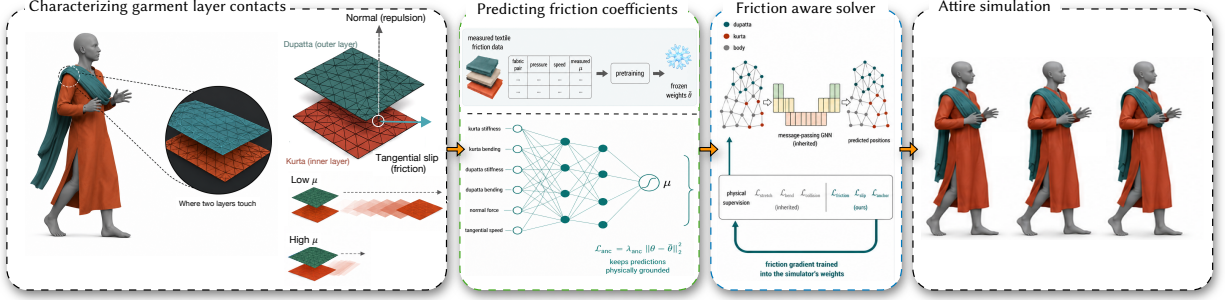


Figure 1: MDSim Architecture. Active contacts are detected between multi-layer garment surfaces. The learned material-conditioned MLP predictor estimates the per-contact friction coefficient μ_i using fabric property stiffness, normal forces, and speed. The resulting loss, regularized by an anchoring constraint, supervises GNN dynamics.

Abstract

In this work, we propose MDSim, a differentiable inter-garment friction framework for neural simulation of multi-layer draped garments. Draped attire (e.g., dupattas, sarees) has no closed geometric anchor and stays on the body through friction against the layer beneath it alone, yet existing learned cloth simulators primarily model repulsive normal contact, leaving such a layer with nothing resisting tangential motion and free to slide off. To address this, MDSim learns physical tangential friction during self-supervised training via a differentiable friction loss that penalizes tangential slip relative to normal force, scaled by a learned, material-conditioned friction predictor pretrained on published textile friction measurements. We additionally isolate a two-garment coupling instability arising from large-area contact between draped layers. Across a range of body motions, MDSim keeps the drape on the shoulder where the unmodified simulator loses it.

CCS Concepts

• **Computing methodologies** → **Physical simulation**; **Neural networks**.

ACM Reference Format:

Anonymous Author(s). 2026. MDSim: Neural Multi-Layer Draped Garment Simulation with Differentiable Friction. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 2 pages. <https://doi.org/XXXXXXX.XXXXXXX>

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Conference acronym 'XX, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/2026/06

<https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Multi-layer draped garments (e.g., dupattas, sarees) lack closed topologies and rely entirely on friction to stay on the body, unlike tailored garments that are pinned or skinned [4–6]. Without tangential forces, draped garments slide off the body (Fig. 2). MDSim integrates physical inter-garment friction into self-supervised GNN training, penalizing slip relative to normal forces, scaled by a material-conditioned friction coefficient μ_i pretrained on textile data. Our main contributions are: (1) **an inter-garment friction loss** for neural cloth simulation, trained end to end rather than applied as a separate physics correction; (2) **a learned, per-contact friction coefficient** conditioned on the two contacting fabrics and pretrained on measured textile data [3, 8], replacing a single hand-chosen constant; (3) **multi-layer drape retention** in a neural simulator, addressing a culturally significant clothing category absent from the literature; and (4) **isolation of a two-garment coupling instability**, in which large-area contact drives the baseline simulator’s own learned dynamics to diverge, independent of friction and relevant to future multi-garment methods.

2 Related Work

Physics solvers resolve friction [1, 2] but are slow. Neural simulators trade accuracy for speed [5], and ContourCraft [4] resolves intersections, but learned contact remains purely repulsive. Parametric datasets [6] exclude draped styles or apply non-physical pin constraints to avoid sliding errors. Textile engineering separately measures fabric-on-fabric friction [3, 8], grounding our predictor, though only via isolated test-rig samples rather than garments in motion. We instead learn differentiable inter-garment friction directly within neural cloth dynamics.

3 Method

We build on ContourCraft [4], a GNN predicting dynamics under repulsion energies (Fig. 1), and introduce a tangential friction loss scaled by a predicted coefficient. Active contacts are detected within



Figure 2: Qualitative results. Top: Vanilla ContourCraft simulator fails to retain the dupatta, leading to severe sliding and eventual separation. Bottom: Our MDSim framework with differentiable friction maintains the draping over the shoulder across the motions tested.

distance ϵ of a face on a different garment, yielding signed distance d and normal $\hat{n}$.

3.1 The inter-garment friction loss

From relative displacement $\Delta \mathbf{r}$ at contact, we remove the normal component to get tangential slip: $\mathbf{s} = \Delta \mathbf{r} - (\Delta \mathbf{r} \cdot \hat{n}) \hat{n}$. Bounded normal force $F_n = (\text{clamp}(\epsilon - d, 0)/\epsilon)^2 \in [0, 1]$ is gradient-detached so friction governs sliding while repulsion governs separation. To model stick-slip, we penalize slip using a pseudo-Huber function $\phi(a) = \sqrt{a^2 + \delta^2} - \delta$ ($\delta = 1$ mm). Summing over N active contacts yields:

$$\mathcal{L}_{\text{IGF}} = \sum_{i=1}^N \mu_i F_{n,i} \phi(\|\mathbf{s}_i\|), \quad (1)$$

where the friction coefficient μ_i is dynamically predicted per contact.

3.2 A learned per-contact friction coefficient

To handle mixed materials, a small MLP g_θ predicts $\mu \in [\mu_{\min}, \mu_{\max}]$ per contact as $\mu = \mu_{\min} + (\mu_{\max} - \mu_{\min})\sigma(g_\theta(\mathbf{f}))$, where $\mathbf{f}$ is an 8D vector of fabric stiffnesses, normal force, and speed. Including relative velocity in $\mathbf{f}$ lets the network model velocity-dependent friction regimes (Stribeck effects), where friction decreases as sliding speed increases. We pretrain the 3-layer MLP (32/16 units, LeakyReLU) offline using MSE loss on published friction measurements for cotton and polyester woven fabrics [3, 8] to model dynamic friction. During GNN training, we set $\lambda_{\text{anc}} = 10^{-4}$ to anchor g_θ to this prior.

3.3 Anchoring and retention

To prevent parameter collapse, the anchoring loss $\mathcal{L}_{\text{anc}} = \|\theta - \bar{\theta}\|^2$ penalizes deviation from the frozen pretraining weights $\bar{\theta}$, preserving the pretrained physical priors during adaptive fine-tuning. We also introduce a retention loss $\mathcal{L}_{\text{ret}}$ to prevent layers from separating completely: it detects a vertex in contact on the previous step but not the current one, penalizing its drift from the last contact position. We set $\mu_{\min} = 0.05$, $\mu_{\max} = 1.5$, and $\epsilon = 3$ mm, weighting

the terms w_{IGF} , w_{anc} , and w_{ret} relative to the baseline (see supplementary material). The full objective is $\mathcal{L} = \mathcal{L}_{\text{base}} + w_{\text{IGF}} \mathcal{L}_{\text{IGF}} + w_{\text{anc}} \mathcal{L}_{\text{anc}} + w_{\text{ret}} \mathcal{L}_{\text{ret}}$, where $\mathcal{L}_{\text{base}}$ denotes all baseline terms.

4 Results and Discussion

Setup. We evaluate on a multi-layer outfit: an inner kurta (5,106 vertices) and an outer, unpinned dupatta (4,958 vertices) draped over the shoulders of an SMPL body [7]. The dupatta is initialized with a small gap to avoid a frame-zero singularity; the garments settle on a static body before a range of dynamic motions begins, yielding an intersection-free start. We compare vanilla ContourCraft against MDSim with identical parameters.

Coupling Instability. While the kurta alone remains stable, adding the dupatta drives smooth exponential divergence, with vertex displacements growing to several meters within a single rollout (one uninterrupted sequence of predicted frames). Draped attire involves large, sustained contact areas where the two layers inject numerical energy that GNNs (tuned for loosely coupled, stitched garments) fail to dissipate. This instability worsens during rapid rotations or sudden speed changes, where the lack of friction lets layers slide freely, accumulating high-frequency vibrations that explode the GNN solver.

Qualitative Behavior. Without friction, the dupatta slides off the shoulders within the first second; MDSim resolves this, stabilizing dynamics and keeping the dupatta draped across the motions tested (Fig. 2).

5 Conclusion

We have shown that incorporating differentiable friction stabilizes GNN dynamics on multi-layer draped garments by preventing unphysical sliding, replacing repulsion-only contact resolution with a friction-aware framework that unlocks stable long-term simulation of unpinned attire. Future work will focus on quantitative evaluation across fabric types (silk, cotton, wool) to test the predictor’s generalization under varying textile tribology, and on extending this approach to complex multi-layered garments (e.g., sarees with pleats or layered ghagras), with potential applications in real-time spatial computing and digital human animation.

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